

Cloud Computing

Chapter 4 – Web OS

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1 Web OS

2 Web OS Architecture

3 Comparison

Introduction to Web OS

- A **Web Operating System (Web OS)** is a user interface that allows users to access applications via the Internet.
- It does **not directly control hardware** like traditional OS.
- Works on top of existing OS (Windows, Linux, macOS).
- Provides access to applications stored on **remote servers**.

Traditional OS vs Web OS

- Traditional OS interacts directly with hardware.
- Web OS provides access to cloud-based applications.
- Traditional apps run locally, Web OS apps run on servers.
- Web OS depends on Internet connectivity.

What Does Web OS Do?

- Provides access to:
 - Email services
 - File management
 - Word processing
 - Spreadsheets
 - Media editing tools
 - Messaging applications
- Applications are stored on **cloud servers**.
- Data is saved in **online databases**.

How Web OS Works

- User accesses Web OS via browser or client software.
- Request is sent to a **control node (server)**.
- Server connects to application servers/database.
- Results are sent back to the user.

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Architecture of Web OS

Client → Internet → Control Server → Application Server → Database

- Client: Browser or application
- Control Server: Manages requests
- Application Server: Executes applications
- Database: Stores data

- **AJAX**

- Uses HTML, CSS, JavaScript, XML
- Enables asynchronous communication

- **Flash (Legacy)**

- Used for interactive UI
- Supports streaming content

AJAX Explained

- Sends data in small parts instead of reloading entire page.
- Improves performance and user experience.
- Used in applications like:
 - Gmail
 - Google Calendar

Why Use Web OS?

- No need for powerful hardware
- Access from anywhere
- Easy collaboration
- No software installation required
- Automatic updates

Advantages of Web OS

- Accessibility from any device
- Cross-platform compatibility
- Centralized data storage
- Easy sharing and collaboration
- Reduced hardware dependency

Disadvantages of Web OS

- Requires Internet connection
- Security concerns (data privacy)
- Performance depends on network
- Limited offline functionality

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Web OS vs Cloud OS vs Traditional OS

Feature	Traditional OS	Web OS	Cloud OS
Execution	Local machine	Browser-based	Cloud infrastructure
Hardware Control	Full control	No direct control	Virtualized control
Dependency	Installed locally	Internet required	Cloud platform required
Examples	Windows, Linux	Web desktops	AWS, Azure OS layer
Data Storage	Local disk	Remote servers	Distributed cloud storage
Scalability	Limited	Moderate	Highly scalable
Performance	High (local)	Network dependent	High (distributed)
Accessibility	Device-specific	Anywhere	Anywhere

Web OS vs Portal

- **Web OS**

- Mimics desktop OS
- Provides application environment

- **Portal**

- Provides access to services
- Does not simulate OS behavior

Real-Life Examples

- Google Docs (online word processor)
- Gmail (web-based email)
- Dropbox (cloud storage)
- Online IDEs (Code editors)

Use Case: Collaboration

- Multiple users work on same file
- Changes saved in real-time
- No need to maintain multiple copies
- Example: Team editing document online

Security Concerns

- Data stored on third-party servers
- Risk of hacking or data breach
- Requires strong encryption and authentication

Future of Web OS

- Increasing adoption of cloud computing
- Integration with AI and automation
- Improved security mechanisms
- Replacement of traditional desktop apps (partial)

Conclusion

- Web OS provides a cloud-based computing experience
- Enables access to applications from anywhere
- Improves collaboration and scalability
- Still evolving with security and performance challenges